MANAGEMENT INFORMATION SYSTEM

UNIT – I

Meaning and Role of MIS

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Unit at a glance

1. Introduction, Definition & Characteristics of MIS **2.** Components of MIS **3.** Nature & Scope of MIS **4.** MIS Organization within the Company **5.** Management, Organizational Theory & Systems Approach **6.** System & Basic System Concepts, Types of Systems **7.** Information System – Definition, Characteristics, Types of Information, Role in Decision-Making **8.** Sub-systems of an Information System: EDP and MIS **9.** Levels of Management – TPS / MIS / DSS

1 Meaning and Role of MIS

1.1 Introduction to MIS

A **Management Information System (MIS)** is a computer-based system that collects data from inside and outside an organisation, processes it, and gives managers the right information at the right time to plan, control and take decisions.

Word	Meaning
Management	The people who plan, organise, direct and control the organisation's work
Information	Processed, meaningful data that is useful for decision-making
System	A group of related parts working together to reach a common goal

Definition

MIS is an **integrated user–machine system** that provides information to support the operations, management and decision-making functions of an organisation. (G. B. Davis)

In simple words: **MIS = collecting data → processing it → delivering useful information to managers.**

1.2 Role (Importance) of MIS

- Gives managers **accurate and timely information** for decisions.
- Helps in **planning and forecasting** by giving past and present data.
- Helps in **control** – compares actual performance with targets.
- Improves **coordination** between departments through shared information.
- Reduces **cost and time** by automating routine reporting.
- Gives a **competitive advantage** through faster, better decisions.

1.3 Characteristics of MIS

Characteristic	Explanation
Management-oriented	Designed around what managers need to plan and control
Integrated	Combines data from all functional areas (finance, marketing, production, HR)
Common database	One shared database avoids duplication and keeps data consistent
Computer-based	Uses hardware and software for fast and accurate processing
Timely & accurate	Information is available when needed and is free from errors
Flexible	Can be changed as business needs change
Future-oriented	Helps in forecasting and planning, not just reporting the past
Exception reporting	Highlights only unusual deviations so managers save time
Sub-system based	Large system is divided into smaller manageable sub-systems
Feedback & control	Results are checked and corrected regularly

2 Components of MIS

An MIS works only when five basic components operate together.

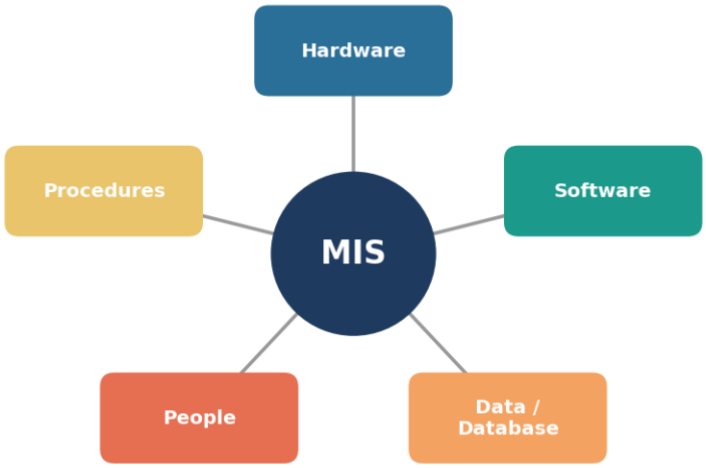


Fig 1: Components of MIS

Component	Description	Example
Hardware	Physical devices used to input, process, store and output data	Computers, servers, printers, scanners
Software	Programs that control the hardware and process data	Operating system, DBMS, ERP, spreadsheets
Data / Database	Raw facts and the organised collection in which they are stored	Sales records, customer and stock data
People	Those who design, operate and use the system	Managers, analysts, programmers, operators

Component	Description	Example
Procedures	Rules and instructions for using and running the system	Backup rules, data-entry steps, security policy

3 Nature and Scope of MIS

3.1 Nature of MIS

- **Information-based** – its main product is information, not just data.
- **Man–machine system** – humans and computers work together.
- **Decision-support tool** – supports but does not replace the manager.
- **Multi-disciplinary** – draws on management, computer science, accounting and behavioural science.
- **Evolving** – grows continuously with technology and business needs.
- **Planned** – must be carefully designed to match organisational goals.

3.2 Scope of MIS

Area	What it covers
Organisational scope	Used by all levels (top, middle, lower) and all departments
Functional scope	Marketing (sales analysis), Finance (budgets), Production (planning), HR (payroll, attendance), Inventory
Technical scope	Hardware, software, databases, networks, security
Managerial scope	Supports planning, organising, directing and controlling
Growing scope	E-commerce, cloud, mobile access, analytics and AI in modern organisations

Advantages of MIS	Limitations of MIS
Better, faster decisions	High cost of setup and maintenance
Improved control and coordination	Quality depends on the quality of data entered
Saves time and effort	Cannot replace human judgement or handle unstructured problems alone
Competitive advantage	Staff resistance and training needs

4 MIS Organization within the Company

The **MIS department** is the unit that builds, runs and maintains the information system. It should report to **top management** so that it gets authority, funds and a company-wide view.

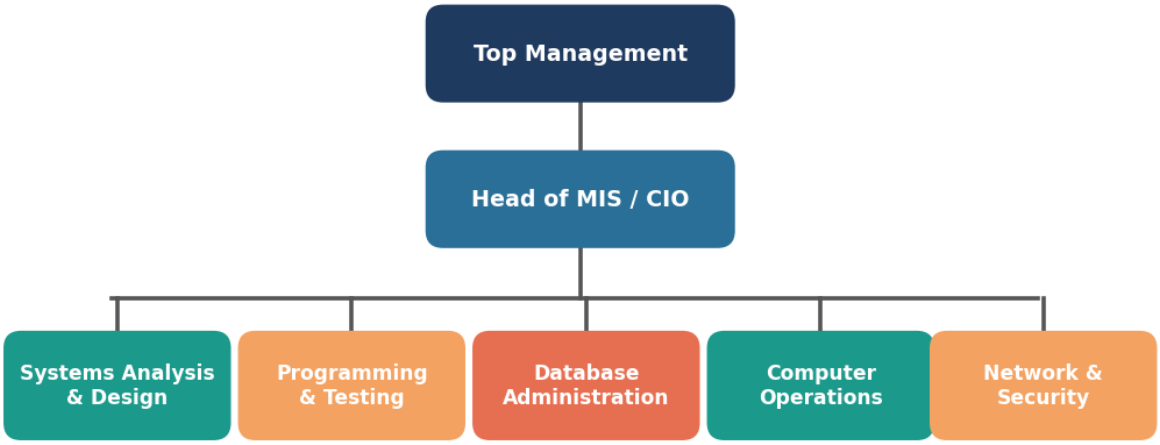


Fig 2: Typical structure of an MIS department

Function	Work done
Systems analysis & design	Studies user needs and designs new systems
Programming & testing	Writes, tests and updates application programs
Database administration	Controls the database, access rights and backups
Computer operations	Runs the systems daily and handles output
Network & security	Maintains networks and protects data from misuse

Centralised MIS	Decentralised MIS
One MIS department serves the whole company	Each department has its own MIS staff and systems
Standard rules, lower cost, better control	Quick response to local needs, but duplication possible

5 Management, Organizational Theory & Systems Approach

5.1 Management

Management is the process of getting work done through people by using resources efficiently to reach organisational goals. Its main functions are:

1. **Planning** – deciding goals and the way to reach them.
2. **Organising** – arranging people and resources.
3. **Staffing** – recruiting and developing people.
4. **Directing** – guiding, motivating and leading.
5. **Controlling** – comparing results with plans and correcting gaps.

MIS supports every function by supplying the information each one needs.

5.2 Development of Organization Theory

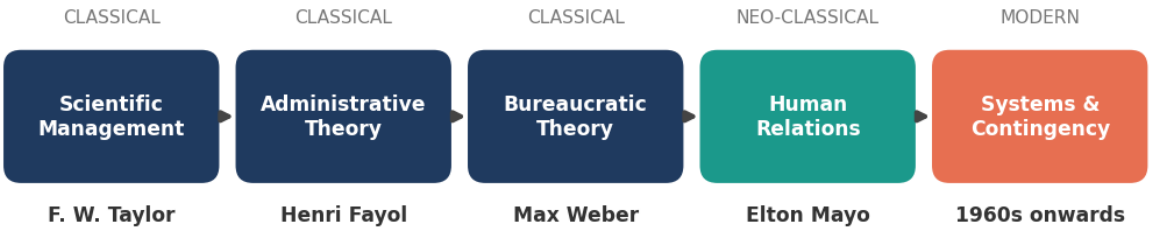


Fig 3: Development of organisation theory

Theory	Main idea
Scientific Management (F. W. Taylor)	Find the "one best way" of doing a job; focus on efficiency and productivity
Administrative Theory (Henri Fayol)	Principles and functions of management applicable at the top level
Bureaucratic Theory (Max Weber)	Clear rules, hierarchy and specialisation make an organisation efficient
Human Relations (Elton Mayo)	Workers' feelings, groups and motivation affect productivity
Systems & Contingency approach	Organisation is an open system; the best method depends on the situation

5.3 Management & Organizational Behavior

Organizational behavior (OB) studies how individuals and groups act inside an organisation. It matters for MIS because a system succeeds only if people **accept and use** it. Motivation, communication, leadership and resistance to change all affect how well an MIS works.

5.4 Information

Data	Information
Raw, unprocessed facts	Processed data that has meaning
Not directly useful for decisions	Useful for decisions
Example: 45, 60, 52	Example: "Average marks of the class are 52"

5.5 The Systems Approach

The **systems approach** views an organisation as a **whole made of inter-related parts** (departments) that interact with each other and with the outside environment, instead of studying each department separately.

- Looks at the **big picture** – a change in one part affects others.
- Treats the organisation as an **open system** that takes inputs and gives outputs.
- Encourages **coordination** and common goals across departments.
- Forms the **foundation of MIS design**.

6 System and Basic System Concepts

6.1 Introduction to System

A **system** is a set of inter-related elements that work together to achieve a common goal. Examples: the human body, a college, a computer, a bank.

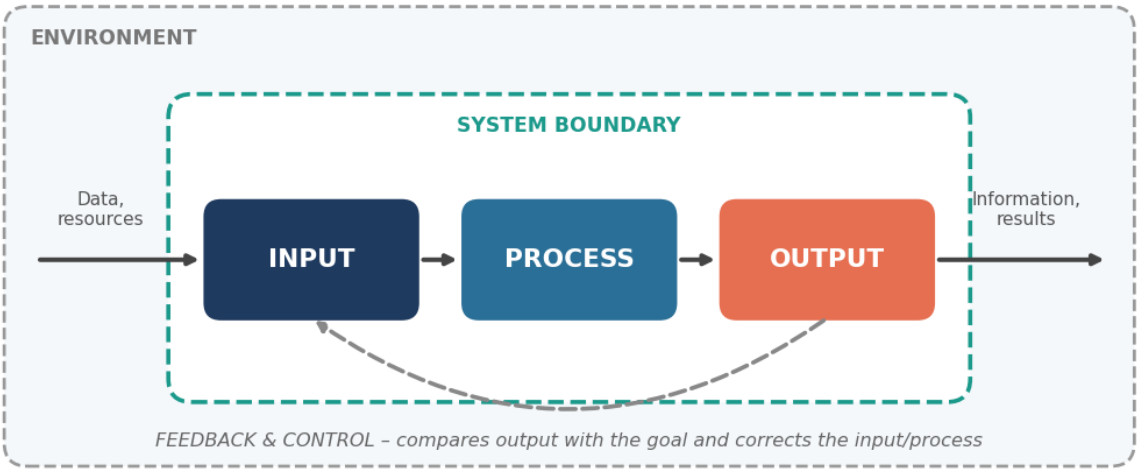


Fig 4: Basic model of a system

Element	Meaning
Input	Data or resources entering the system
Process	Activity that converts input into output
Output	Result produced by the system
Feedback	Information about output sent back to adjust the system
Control	Action to keep performance on target
Boundary	Limit that separates the system from its environment
Environment	Everything outside the boundary that affects the system
Goal	The purpose the system exists to achieve

Characteristics of a system

Organisation • Interaction • Interdependence • Integration • Central objective – parts are arranged, linked, depend on each other and work for one goal.

6.2 Types of Systems

Basis	Type	Meaning / Example
Existence	Physical	Tangible, can be seen – computer, building
	Abstract	Conceptual, not physical – theory, software model
Interaction with environment	Open	Interacts with environment – business, human
	Closed	No interaction – a sealed chemical experiment

Basis	Type	Meaning / Example
Behaviour	Deterministic	Output is predictable – a program, calculator
	Probabilistic	Output is uncertain – weather, stock market
Origin	Natural	Created by nature – solar system, ecosystem
	Man-made	Created by humans – ERP, bank
Adaptability	Adaptive	Changes with environment – a modern organisation
	Non-adaptive	Cannot change – a simple machine

7 Information System

7.1 Definition & Characteristics

An **information system (IS)** is an organised combination of people, hardware, software, data and procedures that **collects, processes, stores and distributes information** to support operations, management and decision-making.

- **Purpose-oriented** – serves a clear organisational goal.
- **Input–process–output–feedback** cycle forms its basic working.
- **Accurate, timely and relevant** information delivery.
- **User-friendly** and available to authorised users.
- **Secure** – data is protected from loss and misuse.

7.2 Types of Information

Basis	Types
By management level	Strategic (top level, long term) • Tactical (middle level, medium term) • Operational (lower level, day-to-day)
By source	Internal (sales, stock, accounts) • External (market, competitors, government rules)
By time	Historical (past) • Current (present) • Predictive (future forecasts)
By nature	Formal (reports, records) • Informal (discussions, rumours)
By use	Planning • Control • Knowledge information

> ☑ **Qualities of good information**
> **Accurate • Timely • Relevant • Complete • Concise • Consistent • Understandable** – information that lacks these can lead to wrong decisions.

7.3 Role of Information in Decision-Making

- Helps in **identifying the problem** clearly.
- Gives facts to **develop and compare alternatives**.
- **Reduces uncertainty and risk** in the final choice.
- Allows **monitoring of results** after the decision is applied.
- Supports **planning, control and coordination** at all levels.

Data → Information → Decision → Action → Feedback

8 Sub-Systems of an Information System: EDP and MIS

An information system has two main sub-systems – **EDP** (Electronic Data Processing) and **MIS**. EDP prepares the data and MIS turns it into management information.

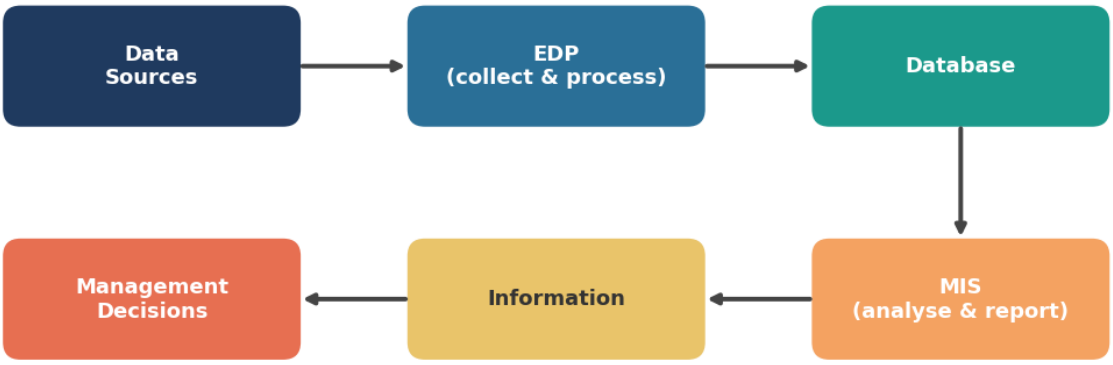


Fig 5: EDP and MIS in an information system

Basis	EDP	MIS
Meaning	Processing of day-to-day transaction data by computers	Producing information for managerial decisions
Focus	Data and transactions	Information and decisions
Level	Operational (clerical) level	Middle and top management
Output	Bills, payroll, ledgers, records	Summary, exception and analytical reports
Example	Preparing monthly salary slips	Report showing salary cost trend by department
Relation	Supplies data to MIS	Uses EDP's data to create information

9 Levels of Management and Systems – TPS / MIS / DSS

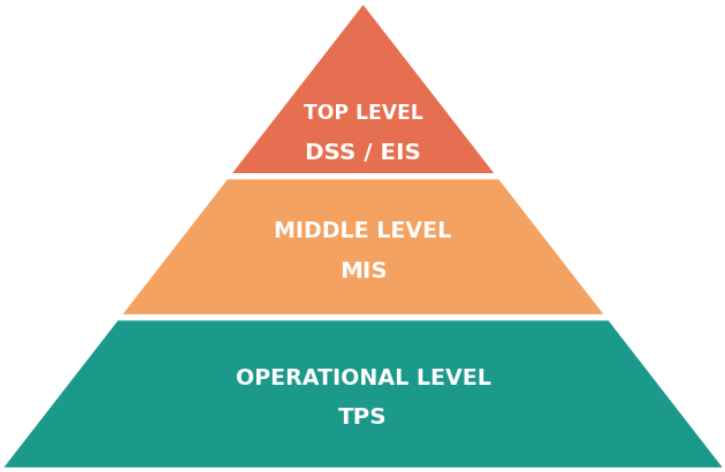


Fig 6: Management levels and the systems that serve them

System	Level	Purpose	Example
TPS – Transaction Processing System	Operational	Records and processes day-to-day transactions quickly and accurately	Billing, ticket booking, payroll
MIS – Management Information System	Middle	Converts TPS data into summary and exception reports for control and planning	Monthly sales report
DSS – Decision Support System	Top / senior	Uses data and models to analyse alternatives for semi-structured and unstructured problems	What-if analysis for pricing

Management level	Role	Information type
Top (Strategic)	Sets goals, long-term plans	External, summarised, future-oriented
Middle (Tactical)	Implements plans, allocates resources	Mostly internal, periodic, summarised
Lower (Operational)	Supervises daily work	Detailed, current, internal

★ Quick Revision & Exam Questions

Key points in one glance

- **MIS** = system that gives managers timely, accurate information for planning, control and decisions.
- **Components:** Hardware, Software, Data, People, Procedures.
- **System elements:** Input, Process, Output, Feedback, Control, Boundary, Environment.
- **EDP** processes data; **MIS** produces information. **TPS → MIS → DSS** serve operational → middle → top levels.
- **Org theory:** Taylor → Fayol → Weber → Mayo → Systems/Contingency.

Likely questions

1. Define MIS and explain its characteristics.
2. Explain the components of MIS with a diagram.
3. Discuss the nature and scope of MIS.
4. How is the MIS department organised within a company?
5. Explain the development of organisation theory and the systems approach.
6. What is a system? Explain basic system concepts and types of systems.
7. What is an information system? Explain the types of information and its role in decision-making.
8. Differentiate between EDP and MIS.
9. Explain TPS, MIS and DSS with respect to the levels of management.